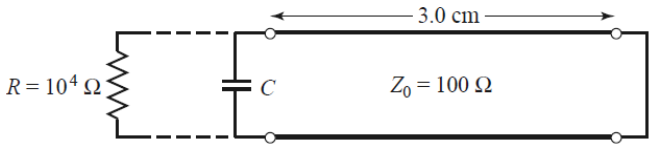
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EXERCISE SESSION 9 (15.01.2021)

Problem 9.1: Basics of resonator

A resonator is constructed from a 3.0 cm length of 100 Ω air-filled coaxial line, shorted at one end and terminated with a capacitor at the other end, as shown below.



(a) Determine the capacitor value to achieve the lowest order resonance at 6.0 GHz.
(b) Now assume that loss is introduced by placing a 10,000 Ω resistor in parallel with the capacitor. Calculate the unloaded *Q*.

Problem 9.2: Coaxial resonator

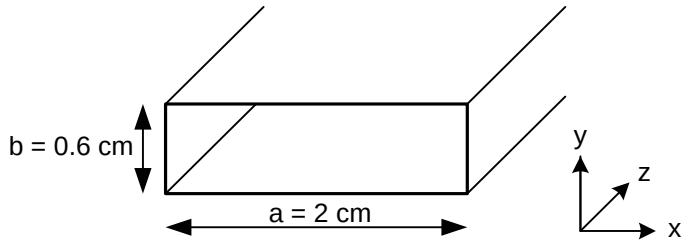
The Q-factor of a quarter-wavelength long TEM-transmission line resonator is approximately given by $Q = \frac{\beta}{2\alpha}$. For a quarter-wavelength long coaxial line, $Z_\mathrm{L} = 16\Omega$, $\epsilon = 27$, $\tan\delta = 0.01$, $\sigma = 20 \cdot 10^6 \frac{\mathrm{S}}{\mathrm{m}}$, inner conductor $d = 2$ mm, calculate dimensions and Q for a resonance frequency of 2 GHz.

Problem 9.3: Parallel plate resonator

A parallel plate waveguide is filled with dielectric material of dielectric constant 100. A short section of this waveguide is now terminated at both ends with electric and/or magnetic boundaries, forming a resonator. Give the shortest possible length of this waveguide section for a resonance frequency of 1 GHz (hint: choose suitable boundaries for the two ends of the waveguide section).

Problem 9.4: Rectangular waveguide resonator and waveguide feed

An RF resonator for a filter is to be fabricated using a vacuum-filled rectangular waveguide section. Consider the conducting walls as perfect electric conductors. The cross-section dimensions are a=2 cm; b=0.6 cm.



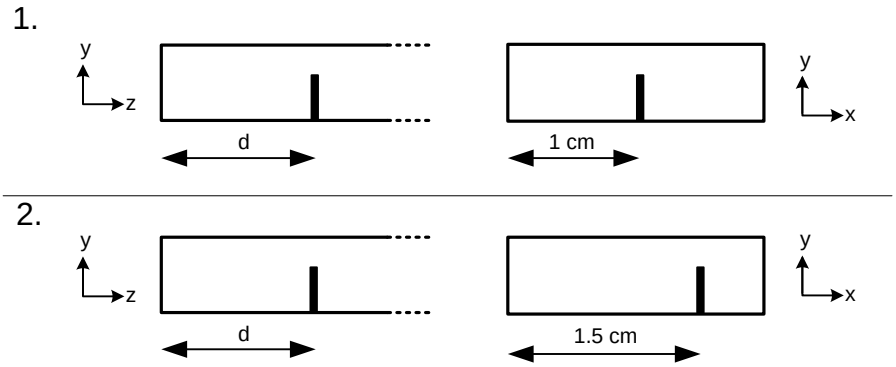
a) Determine the cutoff frequencies and designation of the first 4 modes of the waveguide

b) In which of these modes is possible for a narrowband RF signal of 20 GHz to propagate through the waveguide?

c) A finite section of this waveguide is shorted at both ends to fabricate the resonator. Calculate its physical length for a TE_{203} mode at the desired frequency of 20 GHz (Hint: calculate first the guided wavelength of the TE_{20} mode of the waveguide)

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d) From the following two alternatives, for which the longitudinal and transversal cross-section of the waveguide are sketched, choose the proper way to feed this resonator with a coaxial feed and explain why.



e) Which is the optimal distance d (from one of the shorted walls to the probe) at which the feed should be placed? At which distance d coupling to the resonator does not happen at all?

Short questions session 9

a) What is the phase shift between H and E transversal fields in a standing wave? What is the phase shift of these in a propagating wave?

b) Consider a rectangular waveguide propagating the TE_{10} mode and with a λ_g of 5 cm at a frequency f_0. A resonator is fabricated by shorting both open ends of the waveguide. What should be the length of the resonator to have the TE_{105} resonance mode at the frequency f_0?

c) The resonance frequency of the TM_{110} mode of a rectangular cavity is 3 GHz. What would be the resonance frequency if the length is doubled? (consider the length in z-axis). What is the cutoff frequency of the TM_{11} mode of the waveguide the resonator was designed from?